

| Field Name             | Data Type        | Key Type                           | Description                                                  |
|------------------------|------------------|------------------------------------|--------------------------------------------------------------|
| <b>school_profiles</b> |                  |                                    |                                                              |
| school_id              | VARCHAR(20)      | PK, NN                             | Official DepEd school identifier.                            |
| school_name            | VARCHAR(120)     | NN                                 | Official name of the school.                                 |
| contact_number         | VARCHAR(20)      |                                    | School contact number.                                       |
| email                  | VARCHAR(120)     |                                    | Official school email address.                               |
| created_at             | TIMESTAMP        | NN, DEFAULT CURRENT_TIMESTAMP      | Record creation timestamp.                                   |
| <b>genders</b>         |                  |                                    |                                                              |
| gender_id              | TINYINT UNSIGNED | PK, AI                             | Unique gender identifier.                                    |
| gender_name            | VARCHAR(20)      | UQ, NN                             | Display value such as Male or Female.                        |
| description            | VARCHAR(100)     |                                    | Optional description.                                        |
| <b>majors</b>          |                  |                                    |                                                              |
| major_id               | INT UNSIGNED     | PK, AI                             | Unique major identifier.                                     |
| major_name             | VARCHAR(100)     | UQ, NN                             | Teacher major or specialization name.                        |
| <b>roles</b>           |                  |                                    |                                                              |
| role_id                | TINYINT UNSIGNED | PK, AI                             | Unique role identifier.                                      |
| role_name              | VARCHAR(30)      | UQ, NN                             | Role name such as principal or teacher.                      |
| <b>statuses</b>        |                  |                                    |                                                              |
| status_id              | TINYINT UNSIGNED | PK, AI                             | Unique account-status identifier.                            |
| status_name            | VARCHAR(30)      | UQ, NN                             | Status such as pending, active, rejected, or inactive.       |
| eis_active             | BOOLEAN          | NN, DEFAULT TRUE                   | Indicates whether the status may be assigned.                |
| <b>users</b>           |                  |                                    |                                                              |
| user_id                | BIGINT UNSIGNED  | PK, AI                             | Unique user identifier.                                      |
| school_id              | VARCHAR(20)      | FK -> school_profile.school_id, NN | School that owns the account.                                |
| gender_id              | TINYINT UNSIGNED | FK -> genders.gender_id, NN        | Gender lookup reference.                                     |
| major_id               | INT UNSIGNED     | FK -> majors.major_id              | Teacher major or specialization; optional for principal.     |
| role_id                | TINYINT UNSIGNED | FK -> roles.role_id, NN            | Authorization role.                                          |
| status_id              | TINYINT UNSIGNED | FK -> status.status_id, NN         | Account lifecycle status.                                    |
| first_name             | VARCHAR(50)      | NN                                 | First name.                                                  |
| middle_name            | VARCHAR(50)      |                                    | Middle name.                                                 |
| last_name              | VARCHAR(50)      | NN                                 | Last name.                                                   |
| suffix                 | VARCHAR(10)      |                                    | Optional suffix such as Jr. or III.                          |
| birth_date             | DATE             |                                    | Date of birth used to compute age.                           |
| email                  | VARCHAR(120)     | UQ, NN                             | Unique login email.                                          |
| contact_number         | VARCHAR(20)      |                                    | Primary mobile or contact number.                            |
| password_hash          | VARCHAR(255)     | NN                                 | Secure password hash; plain-text passwords are never stored. |

|                       |                                      |                                           |                                                     |
|-----------------------|--------------------------------------|-------------------------------------------|-----------------------------------------------------|
| created_at            | TIMESTAMP                            | NN, DEFAULT<br>CURRENT_TIMESTAMP          | Account creation timestamp.                         |
| updated_at            | TIMESTAMP                            | NN, AUTO UPDATE                           | Last account update timestamp.                      |
| <b>curriculums</b>    |                                      |                                           |                                                     |
| curriculum_id         | INT UNSIGNED                         | PK, AI                                    | Unique curriculum identifier.                       |
| curriculum_name       | VARCHAR(80)                          | NN                                        | Curriculum name, such as MATATAG Curriculum.        |
| version               | VARCHAR(30)                          | NN                                        | Version or release label.                           |
| description           | TEXT                                 |                                           | Curriculum description.                             |
| status                | ENUM('active','inactive','archived') | NN, DEFAULT active                        | Curriculum lifecycle state.                         |
| created_at            | TIMESTAMP                            | NN, DEFAULT<br>CURRENT_TIMESTAMP          | Record creation timestamp.                          |
| <b>academic_years</b> |                                      |                                           |                                                     |
| academic_year_id      | INT UNSIGNED                         | PK, AI                                    | Unique academic-year identifier.                    |
| curriculum_id         | INT UNSIGNED                         | FK -> curriculum.curriculum_id, NN        | Curriculum used for the academic year.              |
| year_name             | VARCHAR(15)                          | UQ, NN                                    | Display value such as 2026-2027.                    |
| start_date            | DATE                                 | NN                                        | Official opening date.                              |
| end_date              | DATE                                 | NN                                        | Official closing date.                              |
| status                | ENUM('planned','active','completed') | NN                                        | Current academic-year state.                        |
| created_at            | TIMESTAMP                            | NN, DEFAULT<br>CURRENT_TIMESTAMP          | Record creation timestamp.                          |
| updated_at            | TIMESTAMP                            | NN, AUTO UPDATE                           | Last update timestamp.                              |
| <b>term_periods</b>   |                                      |                                           |                                                     |
| term_period_id        | INT UNSIGNED                         | PK, AI                                    | Unique term-period identifier.                      |
| academic_year_id      | INT UNSIGNED                         | FK -> academic_years.academic_year_id, NN | Academic year that contains the term.               |
| term_name             | VARCHAR(50)                          | NN                                        | Display name such as First Quarter.                 |
| term_order            | TINYINT UNSIGNED                     | NN                                        | Chronological order within the academic year.       |
| status                | ENUM('planned','active','completed') | NN                                        | Term state.                                         |
| <b>grade_levels</b>   |                                      |                                           |                                                     |
| grade_level_id        | INT                                  | PK                                        | Unique ID for the grade level.                      |
| grade_level_name      | VARCHAR(20)                          |                                           | The name of the level (e.g., 'Grade 7', 'Grade 8'). |
| <b>sections</b>       |                                      |                                           |                                                     |
| section_id            | INT                                  | PK                                        | Unique ID for each section.                         |

|                |             |    |                                                                        |
|----------------|-------------|----|------------------------------------------------------------------------|
| grade_level_id | INT         | FK | Links this section to a specific grade level in the grade_level table. |
| section_name   | VARCHAR(50) |    | The official name of the section (e.g., 'Rizal', 'Bonifacio').         |

## subjects

|              |             |    |                                            |
|--------------|-------------|----|--------------------------------------------|
| subject_id   | INT         | PK | Unique ID for the subject.                 |
| subject_code | VARCHAR(10) |    | Official or internal subject code.         |
| subject_name | VARCHAR(30) |    | The name of the subject (e.g., 'Science'). |

## students

|             |                                                     |                                    |                                             |
|-------------|-----------------------------------------------------|------------------------------------|---------------------------------------------|
| student_id  | BIGINT UNSIGNED                                     | PK, AI                             | Internal learner identifier.                |
| school_id   | VARCHAR(20)                                         | FK -> school_profile.school_id, NN | School that owns the learner record.        |
| gender_id   | TINYINT UNSIGNED                                    | FK -> genders.gender_id, NN        | Gender lookup reference.                    |
| student_lrn | VARCHAR(12)                                         | UQ, NN                             | Official 12-digit Learner Reference Number. |
| first_name  | VARCHAR(50)                                         | NN                                 | First name.                                 |
| middle_name | VARCHAR(50)                                         |                                    | Middle name.                                |
| last_name   | VARCHAR(50)                                         | NN                                 | Last name.                                  |
| suffix      | VARCHAR(10)                                         |                                    | Optional name suffix.                       |
| birth_date  | DATE                                                |                                    | Date of birth.                              |
| status      | ENUM('active','inactive','transferred','graduated') | NN, DEFAULT active                 | Learner record state.                       |
| created_at  | TIMESTAMP                                           | NN, DEFAULT CURRENT_TIMESTAMP      | Record creation timestamp.                  |

## classes

|                  |                 |                                           |                                             |
|------------------|-----------------|-------------------------------------------|---------------------------------------------|
| class_id         | BIGINT UNSIGNED | PK, AI                                    | Unique class-assignment identifier.         |
| academic_year_id | INT UNSIGNED    | FK -> academic_years.academic_year_id, NN | Unique cohort identifier.                   |
| section_id       | INT UNSIGNED    | FK -> users.user_id, NN                   | Academic year of the cohort.                |
| status           | ENUM            | FK -> subjects.subject_id, NN             | Imported section represented by the cohort. |
| created_at       | TIMESTAMP       | FK -> sections.section_id, NN             | active, completed, or archived.             |
| updated_at       | TIMESTAMP       | NN, DEFAULT CURRENT_TIMESTAMP             | Creation timestamp.                         |

## class\_assignments

|                     |                 |        |                               |
|---------------------|-----------------|--------|-------------------------------|
| class_assignment_id | BIGINT UNSIGNED | PK, AI | Unique assignment identifier. |
|---------------------|-----------------|--------|-------------------------------|

|                 |                 |                               |                                 |
|-----------------|-----------------|-------------------------------|---------------------------------|
| class_id        | BIGINT UNSIGNED | FK -> class, NN               | Section/year cohort.            |
| user_id         | BIGINT UNSIGNED | FK -> user, NN                | Assigned teacher.               |
| subject_id      | INT UNSIGNED    | FK -> subject, NN             | Assigned subject.               |
| assignment_role | ENUM            | NN, DEFAULT primary           | primary or co_teacher.          |
| assigned_at     | TIMESTAMP       | NN, DEFAULT CURRENT_TIMESTAMP | Assignment timestamp.           |
| status          | ENUM            | NN, DEFAULT active            | active, completed, or archived. |
| created_at      | TIMESTAMP       | NN, DEFAULT CURRENT_TIMESTAMP | Creation timestamp.             |
| updated_at      | TIMESTAMP       | NN, AUTO UPDATE               | Last update timestamp.          |

## class\_lists

|               |                 |                   |                                     |
|---------------|-----------------|-------------------|-------------------------------------|
| class_list_id | BIGINT UNSIGNED | PK, AI            | Unique class-membership identifier. |
| class_id      | BIGINT UNSIGNED | FK -> class, NN   | Cohort containing the learner.      |
| student_id    | BIGINT UNSIGNED | FK -> student, NN | Learner in the cohort.              |

## root\_tags

|               |                           |                                    |                                         |
|---------------|---------------------------|------------------------------------|-----------------------------------------|
| root_tag_id   | INT UNSIGNED              | PK, AI                             | Unique root-tag identifier.             |
| curriculum_id | INT UNSIGNED              | FK -> curriculum.curriculum_id, NN | Curriculum that defines the competency. |
| root_tag_name | VARCHAR(100)              | UQ, NN                             | Broad topic name.                       |
| description   | TEXT                      |                                    | Definition or scope of the topic.       |
| status        | ENUM('active','inactive') | NN, DEFAULT active                 | Availability state.                     |
| created_at    | TIMESTAMP                 | NN, DEFAULT CURRENT_TIMESTAMP      | Record creation timestamp.              |

## competency\_tags

|                 |                 |                                 |                                         |
|-----------------|-----------------|---------------------------------|-----------------------------------------|
| competency_id   | BIGINT UNSIGNED | PK, AI                          | Unique competency identifier.           |
| root_tag_id     | INT UNSIGNED    | FK -> root_tags.root_tag_id, NN | Broad analytical category.              |
| competency_name | VARCHAR(255)    | NN                              | Official competency statement or label. |

## skills

|                |                 |                                         |                                                |
|----------------|-----------------|-----------------------------------------|------------------------------------------------|
| skill_id       | BIGINT UNSIGNED | PK, AI                                  | Unique skill-context identifier.               |
| competency_id  | BIGINT UNSIGNED | FK -> competency_tags.competency_id, NN | Competency represented by the skill.           |
| term_period_id | INT UNSIGNED    | FK -> term_period.term_period_id, NN    | Term in which the skill is taught or assessed. |
| grade_level_id | INT UNSIGNED    | FK -> grade_levels.grade_level_id, NN   | Target grade level.                            |
| subject_id     | INT UNSIGNED    | FK -> subjects.subject_id, NN           | Target subject.                                |
| root_tag_id    | INT UNSIGNED    | FK -> root_tags.root_tag_id, NN         | Broad analytical category.                     |
| curriculum_id  | INT UNSIGNED    | FK -> curriculum.curriculum_id, NN      | Curriculum that defines the competency.        |

## interventions

|                   |                                                              |                           |                                      |
|-------------------|--------------------------------------------------------------|---------------------------|--------------------------------------|
| intervention_id   | BIGINT UNSIGNED                                              | PK, AI                    | Unique intervention identifier.      |
| skill_id          | BIGINT UNSIGNED                                              | FK -> skills.skill_id, NN | Skill addressed by the intervention. |
| intervention_type | ENUM('remediation','review','practice','enrichment','other') | NN                        | Type of instructional support.       |
| description       | TEXT                                                         | NN                        | Detailed intervention guidance.      |

|                    |                                                      |                                      |                                                    |
|--------------------|------------------------------------------------------|--------------------------------------|----------------------------------------------------|
| <b>tests</b>       |                                                      |                                      |                                                    |
| test_id            | BIGINT UNSIGNED                                      | PK, AI                               | Unique assessment identifier.                      |
| class_id           | BIGINT UNSIGNED                                      | FK -> class.class_id, NN             | Class for which the assessment is created.         |
| term_period_id     | INT UNSIGNED                                         | FK -> term_period.term_period_id, NN | Term period of the assessment.                     |
| test_name          | VARCHAR(120)                                         | NN                                   | Assessment title.                                  |
| test_type          | ENUM('quiz','exam','diagnostic','long_test','other') | NN                                   | Assessment category.                               |
| test_date          | DATE                                                 | NN                                   | Date administered or scheduled.                    |
| instructions       | TEXT                                                 |                                      | General instructions.                              |
| total_items        | INT UNSIGNED                                         | NN                                   | Total question count across all parts.             |
| status             | ENUM('draft','active','completed','archived')        | NN, DEFAULT draft                    | Assessment lifecycle state.                        |
| created_at         | TIMESTAMP                                            | NN, DEFAULT CURRENT_TIMESTAMP        | Creation timestamp.                                |
| updated_at         | TIMESTAMP                                            | NN, AUTO UPDATE                      | Last update timestamp.                             |
| <b>test_parts</b>  |                                                      |                                      |                                                    |
| test_part_id       | BIGINT UNSIGNED                                      | PK, AI                               | Unique test-part identifier.                       |
| test_id            | BIGINT UNSIGNED                                      | FK -> tests.test_id, NN              | Parent assessment.                                 |
| part_order         | SMALLINT UNSIGNED                                    | NN                                   | Display and processing order.                      |
| part_name          | VARCHAR(80)                                          | NN                                   | Label such as Part I.                              |
| part_type          | VARCHAR(30)                                          | NN, DEFAULT multiple_choice          | Question format; current scope is multiple choice. |
| number_of_items    | SMALLINT UNSIGNED                                    | NN                                   | Number of questions in the part.                   |
| points_per_item    | DECIMAL(5,2)                                         | NN, DEFAULT 1.00                     | Default score for each correct answer.             |
| <b>questions</b>   |                                                      |                                      |                                                    |
| question_id        | BIGINT UNSIGNED                                      | PK, AI                               | Unique question identifier.                        |
| test_part_id       | BIGINT UNSIGNED                                      | FK -> test_part.test_part_id, NN     | Test part containing the question.                 |
| item_number        | INT UNSIGNED                                         | NN                                   | Question number within the part.                   |
| question_text      | TEXT                                                 | NN                                   | Actual test question.                              |
| option_a           | VARCHAR(255)                                         | NN                                   | Text for option A.                                 |
| option_b           | VARCHAR(255)                                         | NN                                   | Text for option B.                                 |
| option_c           | VARCHAR(255)                                         | NN                                   | Text for option C.                                 |
| option_d           | VARCHAR(255)                                         | NN                                   | Text for option D.                                 |
| option_e           | VARCHAR(255)                                         |                                      | Optional option E.                                 |
| <b>answer_keys</b> |                                                      |                                      |                                                    |
| answer_key_id      | BIGINT UNSIGNED                                      | PK, AI                               | Unique answer-key identifier.                      |
| question_id        | BIGINT UNSIGNED                                      | FK -> questions.question_id, UQ, NN  | Question answered by this key.                     |
| correct_option     | CHAR(1)                                              | NN, CK A-E                           | Correct shading option.                            |
| <b>mappings</b>    |                                                      |                                      |                                                    |
| mapping_id         | BIGINT UNSIGNED                                      | PK, AI                               | Unique mapping identifier.                         |
| question_id        | BIGINT UNSIGNED                                      | FK -> questions.question_id, NN      | Question being classified.                         |
| skill_id           | BIGINT UNSIGNED                                      | FK -> skills.skill_id, NN            | Skill measured by the question.                    |

|                             |                                                  |                                        |                                                          |
|-----------------------------|--------------------------------------------------|----------------------------------------|----------------------------------------------------------|
| start_item_no               | INT UNSIGNED                                     | NN                                     | First item number included in the mapped range.          |
| end_item_no                 | INT UNSIGNED                                     | NN                                     | Last item number included in the mapped range.           |
| <b>test_results</b>         |                                                  |                                        |                                                          |
| test_result_id              | BIGINT UNSIGNED                                  | PK, AI                                 | Central result identifier.                               |
| result_uuid                 | CHAR(36)                                         | UQ, NN                                 | Offline-generated idempotency identifier.                |
| test_id                     | BIGINT UNSIGNED                                  | FK -> test, NN                         | Assessment taken.                                        |
| class_list_id               | BIGINT UNSIGNED                                  | FK -> class_list, NN                   | Learner membership.                                      |
| scan_session_id             | BIGINT UNSIGNED                                  | FK -> scan_session, UQ                 | Verified OMR source; nullable for approved manual entry. |
| attempt_number              | SMALLINT UNSIGNED                                | NN, DEFAULT 1                          | Attempt sequence.                                        |
| total_score                 | DECIMAL(8,2)                                     | NN                                     | Sum of verified points earned.                           |
| max_score                   | DECIMAL(8,2)                                     | NN                                     | Maximum possible points snapshot.                        |
| items_evaluated             | INT UNSIGNED                                     | NN                                     | Count of evaluated questions.                            |
| checked_at                  | TIMESTAMP                                        | NN                                     | Teacher verification/check completion time.              |
| <b>student_answers</b>      |                                                  |                                        |                                                          |
| student_answer_id           | BIGINT UNSIGNED                                  | PK, AI                                 | Unique answer-record identifier.                         |
| test_result_id              | BIGINT UNSIGNED                                  | FK -> test_results.test_result_id, NN  | Overall student attempt.                                 |
| question_id                 | BIGINT UNSIGNED                                  | FK -> questions.question_id, NN        | Question answered.                                       |
| is_correct                  | BOOLEAN                                          | NN                                     | Whether the selected option matched the answer key.      |
| points_earned               | DECIMAL(5,2)                                     | NN, DEFAULT 0                          | Points awarded for the item.                             |
| <b>intervention_results</b> |                                                  |                                        |                                                          |
| intervention_result_id      | BIGINT UNSIGNED                                  | PK, AI                                 | Unique recommendation identifier.                        |
| test_result_id              | BIGINT UNSIGNED                                  | FK -> test_results.test_result_id, NN  | Student result receiving the recommendation.             |
| intervention_id             | BIGINT UNSIGNED                                  | FK -> intervention.intervention_id, NN | Recommended intervention.                                |
| <b>syncs</b>                |                                                  |                                        |                                                          |
| sync_id                     | BIGINT UNSIGNED                                  | PK, AI                                 | Unique synchronization event identifier.                 |
| user_id                     | BIGINT UNSIGNED                                  | FK -> users.user_id, NN                | Teacher who initiated the sync.                          |
| test_id                     | BIGINT UNSIGNED                                  | FK -> tests.test_id                    | Downloaded test involved, when applicable.               |
| test_result_id              | BIGINT UNSIGNED                                  | FK -> test_results.test_result_id      | Uploaded result involved, when applicable.               |
| device_identifier           | VARCHAR(100)                                     |                                        | Mobile device identifier or installation ID.             |
| direction                   | ENUM('download','uploaded')                      | NN                                     | Direction of data transfer.                              |
| sync_status                 | ENUM('pending','in_progress','success','failed') | NN                                     | Outcome or current sync state.                           |
| <b>scan_sessions</b>        |                                                  |                                        |                                                          |
| scan_session_id             | BIGINT UNSIGNED                                  | PK, AI                                 | Central scan-session identifier.                         |
| scan_uuid                   | CHAR(36)                                         | UQ, NN                                 | Offline-generated idempotency identifier.                |

|                       |                 |                        |                                                                                           |
|-----------------------|-----------------|------------------------|-------------------------------------------------------------------------------------------|
| test_id               | BIGINT UNSIGNED | FK -> test, NN         | Assessment printed on the sheet.                                                          |
| class_list_id         | BIGINT UNSIGNED | FK -> class_list, NN   | Learner whose sheet was scanned.                                                          |
| scanned_by_user_id    | BIGINT UNSIGNED | FK -> user, NN         | Teacher operating the scanner.                                                            |
| verified_by_user_id   | BIGINT UNSIGNED | FK -> user             | Teacher who verified detections.                                                          |
| device_identifier     | VARCHAR(100)    |                        | Mobile installation/device identifier.                                                    |
| image_hash            | CHAR(64)        |                        | SHA-256 hash for audit without storing the image.                                         |
| scan_status           | ENUM            | NN                     | captured, detected, review_required, verified, or rejected.                               |
| scanned_at            | TIMESTAMP       | NN                     | Image capture timestamp with timezone in API payload.                                     |
| <b>omr_detections</b> |                 |                        |                                                                                           |
| omr_detection_id      | BIGINT UNSIGNED | PK, AI                 | Unique detection identifier.                                                              |
| scan_session_id       | BIGINT UNSIGNED | FK -> scan_session, NN | Parent scanned sheet.                                                                     |
| question_id           | BIGINT UNSIGNED | FK -> question, NN     | Detected question.                                                                        |
| detected_option       | CHAR(1)         | CK A-E                 | Scanner-detected option; null for blank/ambiguous.                                        |
| confidence_score      | DECIMAL(5,4)    | CK 0-1                 | Detection confidence from 0.0000 to 1.0000.                                               |
| detection_status      | ENUM            | NN                     | detected, blank, multiple, or ambiguous.                                                  |
| raw_mark              | TEXT            | NN                     | store the actual detected darkness/confidence per option, not only final detected option. |
| detected_at           | TIMESTAMP       | NN                     | Detection timestamp.                                                                      |